

Stochastic Modeling and Thermodynamics of RNA Folding: Heat Renaturation in Modified Biosensor Geometry

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Abstract—Surface-based RNA biosensors detect specific transcripts through hybridization between immobilized probes and target strands. Detection depends on the folding of individual probes into secondary structure and on how the probes are packed in the dense brush at the chip surface. Existing thermodynamic frameworks compute the folding free energy from the sequence using a partition-function recursion, but approximate the chain height by a sequence-level proxy because no closed-form expression is available for the height of a self-avoiding folded chain. The present work tests that approximation. Self-avoiding random walks on a two-dimensional lattice produce a per-walk measurement of the chain height and the ladder distance, whose ratio multiplicatively corrects the field-coupling parameter of the partition function, and the resulting free energy is averaged over a bootstrap of geometric samples. The corrected helicity curves agree with the prior model at high temperature and diverge in the cold regime, where the chain responds more strongly to the surface field. The pipeline rests on a single simplification, the statistical independence of chain geometry and base pairing, whose limitations motivate a fully coupled Monte Carlo treatment in future work.

Index Terms—RNA folding, biosensors, self-avoiding random walks, partition function, helicity, polymer brush, statistical independence

I. INTRODUCTION

A. RNA

Ribonucleic acid (RNA) is a single-stranded biomolecule that carries genetic information from DNA to ribosomes for protein synthesis and serves as a structural and catalytic agent in many cellular processes [6]. Structurally, RNA is a linear polymer composed of nucleotides drawn from a four-letter alphabet, namely adenine (A), uracil (U), guanine (G), and cytosine (C). Each nucleotide consists of a ribose sugar, a phosphate group, and one of these four nitrogenous bases. The bases pair with each other through hydrogen bonds following Watson–Crick complementarity. A pairs with U through two hydrogen bonds, and G pairs with C through three.

Unlike double-stranded DNA, RNA is naturally single-stranded, which allows it to fold back on itself. When complementary regions along a single strand come into proximity,

hydrogen bonds form between matching bases, producing localized double-helical regions called *stems*. The unpaired segments between or within stems form *loops*, *hairpins*, and *bulges*. The complete pattern of which bases pair with which is called the chain’s *secondary structure*, and it is the central object whose thermodynamics we model in this work.

Two simplifications are conventional in the statistical-mechanics treatment of RNA folding and are used throughout this paper. First, only secondary structure is modelled. The only piece of structural information that enters the thermodynamic calculation is the list of which bases are paired with which; the full three-dimensional shape of the chain is not part of this representation and is reintroduced separately through the geometric simulation of Section III-H. Second, pseudoknotted structures (in which two base pairs (i, j) and (k, l) cross in the sense $i < k < j < l$) are excluded. The resulting class of nested (planar) secondary structures admits an exact dynamic-programming recursion for the partition function [1], [3], which is the basis of all sequence-based thermodynamic calculations in this work.

B. Biosensors

Biosensors are analytical devices used to detect specific biological molecules (such as DNA, RNA, proteins, or small metabolites) in a sample. They typically combine a sensing surface decorated with biological probe molecules with a transducer that converts a recognition event into a measurable signal. Among the most widely deployed biosensors are nucleic-acid microarrays (chip-based assays), in which probes of known sequence are immobilized on a solid surface and target molecules from solution hybridize to them by sequence complementarity [4].

In an RNA biosensor, short single-stranded RNA probes of known sequence are anchored to the chip surface at high surface density, collectively forming a polymer brush. Each probe is chosen to be the reverse complement of a target transcript of interest, in the sense that the bases of the probe pair with the bases of the target according to the Watson–

Crick rules of Section I-A (A with U, G with C). When a target RNA molecule from solution comes into proximity with a probe of matching sequence, hydrogen bonds form between aligned complementary bases along the length of the two strands, and a stable double-stranded duplex is produced. This sequence-specific pairing of two single strands into a duplex is called *hybridization*, and it is the molecular event that the sensor reads out. The presence of a duplex on a given probe is reported as a detectable signal, and because the probe sequence is known and spatially indexed on the chip, the location of the signal identifies which target transcripts are present in the sample. This technology is foundational to gene-expression profiling, transcriptome analysis, and a wide range of diagnostic platforms [7].

A key engineering challenge in such sensors is that the probes themselves are single-stranded RNA chains and fold like any other RNA molecule. A probe that folds back on itself into a hairpin or stem hides part of its binding region, because the bases that should pair with the target are already paired internally. Capture efficiency then drops sharply. To suppress this effect, many designs introduce a static electric field at the chip surface that energetically penalizes folded configurations of the probe [4]. The strength of this field is characterized by a coupling parameter α , which appears explicitly in the configuration energy of the chain (Section III-B) and directly modulates the equilibrium helicity. There is no universally optimal value of α . A small α leaves the probes folded and inaccessible, while increasing α progressively unfolds them, and the practical operating point is fixed by the design of the specific sensor [4]. Designing α correctly requires an accurate thermodynamic model of the chain, which is the role of the partition-function machinery introduced in Section III.

C. Folding and the present work

The thermodynamic equilibrium between folded and unfolded states of an RNA chain at temperature T is captured by the *helicity* θ , defined as the equilibrium fraction of bases that are paired at temperature T . Helicity depends on the chain sequence, the temperature, and (in the surface geometry of a biosensor) on the magnitude of the applied field α . The curve $\theta(T)$ describes the chain's heat-driven folding and unfolding behaviour, and is the principal observable computed in this work.

The folding calculation in this paper is built on three layers of prior work that we use without modification. The first layer is the dynamic-programming partition-function algorithm of McCaskill [3], which evaluates the equilibrium statistics of nested secondary structures of a given sequence in polynomial time. The second layer is the statistical-mechanics treatment of secondary structures of random RNA sequences due to Bundschuh and Hwa [1], which provides the two-letter sequence representation and the form of the configuration energy that we use throughout. The third layer is a biosensor-oriented formulation that is the subject of ongoing, unpublished work outside the scope of this paper. That framework adds to the configuration energy an electrostatic loop penalty $-\alpha D$

representing the static field at the chip surface, and, because the chain height D of a self-avoiding folded chain has no closed-form expression, substitutes D with the sequence-level proxy $(1 - l)$, where l is the ladder distance of the secondary structure. The substitution $D \rightarrow (1 - l)$, the parameter α coupling the brush to an effective field, and the resulting helicity calculation are all part of this biosensor framework rather than of [1] or [3]. The framework is fully tractable but removes any explicit dependence on the actual three-dimensional configuration of the chain.

The contribution of the present work sits on top of all three layers and is purely on the geometric side of the calculation. The missing geometric information, namely the chain shape and the per-realization ratio $D/(1 - l)$, is recovered by direct numerical simulation of self-avoiding random walks on a lattice. The simulated ratios then multiplicatively correct the field-coupling parameter α in the partition function, and the corrected helicity curve is compared with that of the bare prior model, to test how much the analytical substitution depends on the actual variability of the chain shape. The partition-function machinery, the two-letter sequence model, and the analytical substitution $(1 - l)$ are imported from prior work without modification. The comparison rests on a single explicit simplification, namely the statistical independence of geometry and pairing, whose precise content, locations in the pipeline, and limitations are discussed in Section III-H.

II. METHODOLOGY

A. Random walks

A random walk on a lattice is a stochastic trajectory in which a walker moves from one lattice site to a neighbouring site at each step, with the direction chosen according to a prescribed probability distribution [5]. In the simplest case, the walker is equally likely to step in any of the available directions, regardless of its history. Random walks are a minimal model of diffusion and of the conformations of an idealized flexible polymer. For an idealized flexible polymer in solution, the chain backbone can be represented as a random walk in which each step corresponds to one monomer and the step direction encodes the local orientation of the backbone.

An unrestricted random walk allows the trajectory to revisit sites it has already occupied, so two monomers along the chain can sit on top of each other. This is unphysical for a real polymer, in which two monomers cannot share the same point in space. To model real polymers we need a walk that respects this excluded-volume condition.

B. Self-avoiding random walks

A *self-avoiding random walk* (SAW) is a random walk subject to the constraint that the trajectory never revisits a site it has previously occupied [5]. The constraint reflects the steric repulsion between monomers and is a standard model in polymer physics, particularly well suited for representing single-stranded RNA chains anchored to a surface.

In our simulations each walk represents one polymer of length N on a two-dimensional square lattice, anchored at the

origin to a horizontal surface that represents the biochip. At each step the walker tries a randomly chosen neighbouring site, and if that site has already been visited the move is rejected and a different direction is tried. Walks that become trapped before reaching the target length are discarded. We use a minimum length of $N \geq 100$ and keep only walks that reach it.

For each successful walk we measure two geometric observables. The first is the chain height D , defined as a normalized average vertical distance of the monomers from the anchoring surface,

$$D = \frac{2}{N(N+1)} \sum_{i=1}^N y_i, \quad (1)$$

where y_i is the height of monomer i above the surface. The factor $2/(N+1)$ normalizes D so that a chain that lies flat on the surface gives $D = 0$ and a chain fully extended perpendicular to the surface gives $D = 1$.

The second observable is the ladder distance l . A geometric contact is a pair of monomers identified by the greedy matching procedure of Section III-F, which pairs monomers that sit at unit lattice distance in space and are not chain-adjacent. Once the contacts have been identified, they are stored in a symmetric contact matrix M , with $M_{jk} = M_{kj} = 1$ if monomers j and k are matched and $M_{jk} = 0$ otherwise. The ladder distance is then computed from M by the triple sum

$$l = \frac{2}{N(N+1)} \sum_{i=1}^N \sum_{j=1}^{i-2} \sum_{k=i}^N M_{jk}. \quad (2)$$

The three loops in (2) have a direct geometric reading. The outer index i runs over chain positions and is used as a separator between a “left side” and a “right side” of the chain. The middle index j ranges over monomers strictly to the left of the separator with at least one monomer of gap ($j \leq i-2$). The inner index k ranges over monomers at or to the right of the separator ($k \geq i$). For each such triple, we add the indicator M_{jk} , that is, we count the pair (j, k) exactly when monomers j and k are in contact. The effect of summing over all i is that a fixed contact (j, k) with $j < k$ is counted once for every separator i that falls between them in the sense $j+2 \leq i \leq k$. Each such separator corresponds to one of the $k-j-1$ chain positions lying strictly between j and k , that is, to one of the monomers enclosed in the loop closed off by the contact. The triple sum is therefore equal to a sum over all contacts of the loop length they enclose, with the normalization factor $2/(N(N+1))$ chosen so that l takes values comparable to those of D . A walk with no contacts gives $l = 0$, and a walk with many contacts enclosing long loops gives l close to its maximum. The qualitative picture of l is therefore a normalized total loop content of the chain. Both D and l are purely geometric. D characterizes the brush extension of the chain, while l characterizes the density of loop-forming returns of the chain to its own immediate neighbourhood.

The central derived observable is the ratio

$$r = \frac{D}{1-l}, \quad (3)$$

which compares the simulated chain height D to the analytical substitution $(1-l)$ used in the prior model (Section III-E). A value $r = 1$ would mean the analytical substitution exactly reproduces the simulated geometric height. Deviations from unity quantify the discrepancy between the analytical proxy and the directly simulated configuration. We accumulate r values from independent SAW realizations until 200 valid walks (those reaching $N \geq 100$) have been collected, and these 200 ratios form the sample distribution of geometric correction factors that is handed to the thermodynamic calculation of Section III.

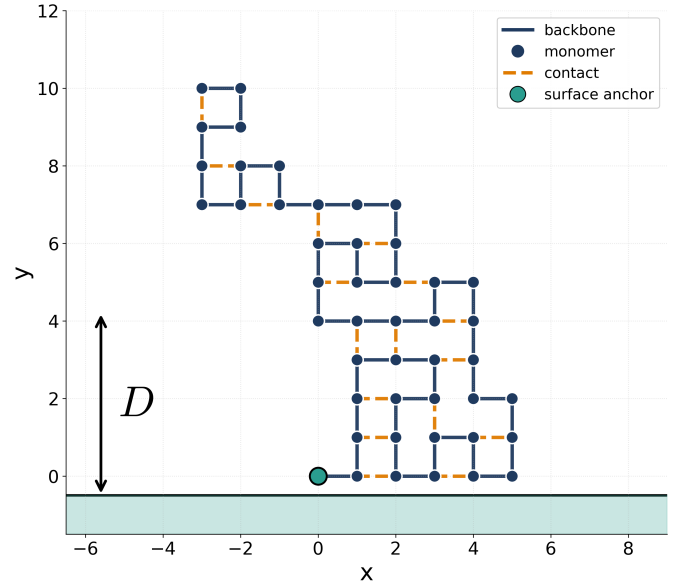


Fig. 1. A single self-avoiding walk anchored to the surface $y = 0$. The chain backbone is drawn as a navy line connecting monomer positions; orange dashed segments mark the geometric contacts identified by the greedy matching procedure of Section III-F. The arrow on the left indicates the chain height D , the normalized average vertical position of the monomers as defined in equation (1). The ladder distance l , defined in equation (2), is a global statistic that counts, for every chain position i and every contact (j, k) that brackets i , a unit of loop content; it is not associated with any single contact in the figure.

III. THERMODYNAMICS

As detailed in the introduction, the thermodynamic machinery used in this section is taken without modification from prior work [1]–[3] together with the biosensor formulation summarized there. The present section reproduces it only as the setting in which the geometric simulation of Section II-B is inserted. The new content of this paper begins at Section III-E (paragraph “Recovery of the geometric chain height”) and is concentrated in Sections III-F and III-H.

A. Statistical-mechanics framework

Because much of the discussion of folding, geometry, and their coupling relies on a small set of statistical-mechanics ideas, we briefly review them here in a form adapted to the present problem.

1) *Microstates and the partition function*: At thermal equilibrium, a system in contact with a heat reservoir at temperature T is described by the canonical ensemble. The system's possible microscopic configurations are labeled by an index s and each microstate carries an energy E_s . The partition function is the sum of Boltzmann weights over all microstates,

$$Z = \sum_s \exp(-\beta E_s), \quad \beta = \frac{1}{k_B T}, \quad (4)$$

where k_B is Boltzmann's constant. The probability of finding the system in microstate s is

$$P(s) = \frac{1}{Z} \exp(-\beta E_s). \quad (5)$$

The Helmholtz free energy is

$$F = -k_B T \ln Z = U - TS, \quad (6)$$

where $U = \langle E \rangle$ is the average energy and S is the Gibbs entropy

$$S = -k_B \sum_s P(s) \ln P(s). \quad (7)$$

At fixed T , equilibrium is the configuration of probabilities $P(s)$ that minimizes F .

In our problem the microstates are secondary structures M of the RNA chain, each characterized by its base-pairing matrix; details of the energy assignment are given in Section III-B.

2) *The heat bath*: A folded RNA chain in solution is in continuous contact with the surrounding water, which acts as a thermal bath at temperature T . The chain itself does not possess a temperature in any independent sense; it inherits the bath's temperature through random energy exchanges. The bath does not steer the chain into any specific configuration; it provides random energy exchanges whose typical size scales with $k_B T$, which is the only parameter the canonical ensemble (5) retains about the bath.

B. Configuration energy

The energy of a folded RNA configuration in the prior thermodynamic framework has two contributions. The first is a sum over base-pair interaction energies along the secondary structure M ,

$$E_{\text{pair}}(M, \{h\}) = \sum_{(i,j) \in M} \varepsilon_{ij} h_i h_j, \quad (8)$$

where ε_{ij} is the pair interaction energy and h_i encodes the nucleotide type at position i (defined more precisely below). The second contribution is an electrostatic penalty arising from the static field generated at the chip surface,

$$E_{\text{field}} = -\alpha D, \quad (9)$$

where α is a phenomenological field-coupling parameter and D is the average geometric height of the chain above the chip. Since D is smaller for folded (more compact) chains, the term

$-\alpha D$ is less favorable for folded configurations, and the field acts to disfavor folding. The total configuration energy is

$$E(M, \{h\}) = \sum_{(i,j) \in M} \varepsilon_{ij} h_i h_j - \alpha D. \quad (10)$$

The challenge with (10) is that D depends on the three-dimensional configuration of the chain, which is not part of the secondary-structure description M . Resolving this mismatch is the subject of Sections III-E and III-H.

C. Two-letter versus four-letter representation

A real RNA chain contains four base types and obeys the Watson-Crick complementarity rules A-U and G-C. A faithful sequence-level model would assign distinct interaction energies to each possible pairing, leading to a 4×4 interaction matrix in which most entries are forbidden. This introduces substantial complexity into the partition-function recursion, and exact analytical treatments become unwieldy.

A standard simplification, used throughout the prior literature and in this work, is the *two-letter* representation, in which each monomer is assigned one of two values, $h_i \in \{-1, +1\}$ [1], [2]. The interaction energy is then written in the compact form

$$\varepsilon_{ij} = \varepsilon_0 + \varepsilon h_i h_j, \quad (11)$$

where ε_0 is a uniform base-contact energy and ε controls the strength of the type-dependent contribution. This corresponds to grouping the four real bases into two effective complementary classes (e.g. $\{A, G\}$ versus $\{U, C\}$), preserving the essential statistical mechanics of complementary pairing in a tractable form.

The corresponding four-letter representation introduces a second sequence variable $g_i \in \{-1, +1\}$ and an additional energy term, yielding

$$\varepsilon_{ij} = \varepsilon_0 + \varepsilon h_i h_j + \varepsilon' g_i g_j, \quad (12)$$

which has been studied in detail in [2] and shown to reproduce the same qualitative thermodynamic features as the two-letter case, including the reentrant melting discussed below. The two-letter abstraction is well-justified for the present work, where we are concerned with generic thermodynamic features (such as the temperature dependence of helicity, the broad shape of the folding-unfolding transition, and the response to external parameters) rather than with sequence-specific structure prediction.

D. Reentrant melting and the role of sequence disorder

When the two-letter model is run with competitive base-pair energies (favorable for unequal bases A-U and unfavorable for equal bases A-A or U-U), the helicity has a non-monotonic temperature dependence, characterized in detail in [2]. The standard picture predicts θ to decrease monotonically with T , reflecting the entropy-energy competition implicit in (6). In a sequence-disordered chain with competing pair energies, however, $\theta(T)$ acquires an additional low-temperature melting branch, producing a maximum in θ at intermediate T and

a corresponding double peak in the specific heat $C_V = -T \partial^2 F / \partial T^2$. The phenomenon is referred to as *reentrant melting* or *cold denaturation* and offers a sequence-disorder-driven mechanism for an experimental observation that has been variously attributed to hydrophobic effects, electrostatics, and hydrogen-bond competition [2].

For the present work, the result of [2] matters in two ways. First, it confirms the two-letter representation as a faithful starting point, since the additional structure observed in the four-letter case (12) is qualitatively the same as in the two-letter case (11), so any generic conclusion drawn at the two-letter level translates to the more realistic alphabet. Second, it shows that the helicity curve $\theta(T)$ is sensitive to small thermodynamic effects, which makes it a meaningful target for our comparison. Since our two pipelines differ only in how they treat geometry, any visible gap between them tells us whether geometric variability matters in the same regime where these subtle effects are most pronounced.

E. Free energy via the McCaskill recursion

The thermodynamic equilibrium of the chain at temperature T is governed by the Helmholtz free energy (6) with the partition function (4), where the index s now ranges over all admissible (nested) secondary structures M . For a chain of length N , the number of distinct nested secondary structures grows exponentially with N , making naive enumeration intractable.

The McCaskill recursion [3] provides an exact dynamic-programming algorithm that computes the partition function in polynomial time. The algorithm assigns to each contiguous sub-segment $[i, j]$ of the chain a partial partition function $Z_{i,j}$, equal to the sum of Boltzmann weights over all nested secondary structures of the sub-chain spanning positions i through j , with the recursion

$$Z_{i,j} = Z_{i,j-1} + \sum_{k=i}^{j-1} Z_{i,k-1} Q_{k,j} Z_{k+1,j-1}, \quad (13)$$

in which $Q_{k,j} = \exp(-\beta \varepsilon_{k,j})$ is the statistical weight of pairing nucleotides k and j . The full partition function is $Z = Z_{1,N}$, and the per-monomer free energy is

$$f = -\frac{1}{N\beta} \ln Z_{1,N}. \quad (14)$$

The prior approximation: The chain height D appearing in the configuration energy (10) is not accessible from a sequence-only calculation, because the recursion (13) does not track three-dimensional positions of monomers. To keep the model tractable, the biosensor framework substituted

$$D \longrightarrow 1 - l, \quad (15)$$

where l is the ladder distance of the secondary structure, expressible directly in terms of the pairing matrix M . The electrostatic term in (10) then becomes $-\alpha(1-l)$, a quantity depending only on M through l and naturally accommodated by (13). The substitution $D \rightarrow (1-l)$ is physically motivated (a chain with more internal contacts tends to be more compact) but it is an analytical approximation, not an exact identity.

Recovery of the geometric chain height: For each realization s , the geometric simulation of Section II-B produces the ratio $r_s = D_s / (1 - l_s)$ defined in (3). This ratio carries the geometric information that the substitution $(1-l)$ alone cannot, since multiplying the substituted electrostatic term by r_s gives

$$-\alpha(1-l)r_s = -\alpha D_s \quad (16)$$

and restores the true chain height D_s in the energy. We therefore replace the bare coupling α by an effective field-coupling

$$\alpha_{\text{eff}} = \alpha \cdot r_s = \alpha \cdot \frac{D_s}{1 - l_s} \quad (17)$$

and average the resulting free energy over the geometric sample,

$$F_{\text{corr}}(\alpha, \beta) = \frac{1}{S} \sum_{s=1}^S F(\alpha \cdot r_s, \beta). \quad (18)$$

Two properties of this construction are worth noting. First, the partition function itself never sees the chain height D directly. It sees the scalar α_{eff} , whose value happens to depend on D through the SAW. The geometric simulation and the partition function therefore communicate through a single number per realization. Second, the partition-function machinery is unchanged. The geometric correction is multiplicative on α and leaves the recursion (13) untouched.

Before (18) is evaluated, the population of r values is rescaled so that $\langle r \rangle = 1$ over the SAW sample. The rescaling ensures that the bare model ($r_s \equiv 1$) and the corrected model use the same mean coupling, so that any difference between their predictions reflects the *variability* of r rather than a shift in its mean. Without this normalization, the average effect could be reproduced trivially by adjusting the value of α in the bare model, and the comparison would degenerate into a parameter-fitting exercise.

The averaging (18) is implemented in a bootstrap fashion. From the normalized population of N_{tot} valid SAW samples, B subsamples of size S are drawn independently and with replacement. For each subsample, the sample-averaged free energy is computed and stored as one bootstrap estimate. The mean across bootstraps provides the central estimate of F_{corr} , while the standard deviation across bootstraps provides an error band used to construct confidence intervals on the helicity curves of Section IV.

F. Structures admitted by each pipeline

The two pipelines used in this work, namely the SAW for geometry and the partition-function recursion for pairing, are based on different algorithms and therefore admit slightly different families of structures. The differences are not physical disagreements but standard simplifications of each model. This subsection enumerates the common RNA loop motifs and, for each, notes which pipeline admits the motif and which does not.

1) *Algorithmic admissibility in the SAW*: The SAW state space is generated by an incremental growth algorithm. At each step, the walker chooses a neighbouring lattice site uniformly at random subject to two constraints. The first constraint is that the new site must lie on or above the anchoring surface ($y \geq 0$), enforced explicitly by skipping any candidate with $y < 0$. The second constraint is that the new site must not coincide with any previously occupied site, enforced by checking against the visited set. Walks that become geometrically trapped before reaching the target length are discarded.

Contacts are then identified post hoc by a greedy matching procedure. For each monomer i in chain order, the procedure inspects its four lattice neighbours, and among those neighbours that are themselves monomers of the same chain with index j satisfying $|i - j| > 1$ and that have not yet been matched, it pairs i with the one of smallest chain index j and marks both as matched. The matching has three structural consequences.

- *One partner per monomer*. A monomer, once matched, is removed from further consideration. Geometric configurations in which several non-chain neighbours of a single monomer are themselves chain monomers are therefore recorded as only one contact.
- *Minimum chain separation* $|i - j| \geq 2$. Chain-adjacent monomers ($|i - j| = 1$) are explicitly excluded because they are connected through the backbone.
- *No nesting check*. The procedure inspects each monomer in isolation, with no test for whether the resulting pair crosses an earlier pair in chain order. Two contacts (i, j) and (k, l) with $i < k < j < l$ (the definition of a pseudoknot) are therefore admitted whenever the geometry produces them.

Hairpin loops, internal loops, bulges, multi-loops, exterior unpaired regions, and pseudoknotted contact patterns are all geometrically realizable on the lattice and all admitted by the matching procedure. The SAW state space is consequently a broad superset of the nested secondary-structure family.

2) *Algorithmic admissibility in the partition function*: The partition-function recursion (13) is also a constructive algorithm, and its admissible state space is fixed by the form of the recurrence. The recurrence assigns to each contiguous sub-segment $[i, j]$ a partial partition function $Z_{i,j}$ and constructs it from three kinds of terms.

- $Z_{i,j-1}$, corresponding to the case in which base j is unpaired.
- $Q_{i,j} Z_{i+1,j-1}$ and the special $i = j - 1$ form $Q_{i,j}$, corresponding to the case in which i pairs with j , enclosing the sub-segment $[i + 1, j - 1]$ that is summed over its own admissible substructures.
- A bifurcation sum $\sum_{k=i+1}^{j-1} Z_{i,k-1} Q_{k,j} Z_{k+1,j-1}$, corresponding to a pair (k, j) splitting $[i, j]$ into two disjoint sub-segments that each fold independently.

The structural constraints embodied in this recurrence are as follows.

- *One pair per base*. Each monomer participates in at most one pair. A sub-segment that has been used to produce $Z_{i+1,j-1}$ (because i and j are now paired) is consumed and cannot pair with anything outside $[i, j]$ in this branch of the recursion.
- *Strict nesting (no pseudoknots)*. The bifurcation term (13) partitions $[i, j]$ into a left segment $[i, k - 1]$ and a right segment $[k + 1, j - 1]$ whose admissible structures are summed independently. There is no term in the recurrence that allows a pair to span across this partition. Consequently, no recurrence path can produce two pairs (i, j) and (k, l) with $i < k < j < l$, so such a configuration is never assembled, and its Boltzmann weight is not included in Z . A pseudoknotted structure has probability zero in this ensemble, and the model treats the chain as if such structures did not exist.
- *Adjacent-base pair admitted*. The special branch $i = j - 1$ admits a pair $(i, i + 1)$ that closes a hairpin loop of zero unpaired bases. Standard McCaskill imposes a minimum loop length $L_{\min} \geq 3$ for steric realism, while the implementation used here corresponds to $L_{\min} = 0$ and is a known simplification.

Hairpin loops, internal loops, bulges, multi-loops, exterior unpaired regions, and stems (stacks of nested pairs) are all assembled by combinations of the three recurrence terms above. Pseudoknotted contact patterns and structures with multiple pairs per base are not assembled by any such combination and therefore have zero Boltzmann weight in Z .

3) *Asymmetry between the two state spaces*: Table I summarizes which standard loop motifs are admitted by each pipeline. The two pipelines coincide on the bulk of the structural inventory (hairpins, internal loops, bulges, multi-loops, exterior loops, and stems) and differ in three categories whose origin is algorithmic.

TABLE I
ADMISSIBILITY OF RNA LOOP MOTIFS IN THE SAW (GEOMETRIC) AND PARTITION-FUNCTION (SEQUENCE) PIPELINES, AS DETERMINED BY INSPECTION OF THE IMPLEMENTATION. ✓ DENOTES ADMITTED; × DENOTES EXCLUDED.

Motif	SAW	Partition fn.
Hairpin loop	✓	✓
Stem (stack of nested pairs)	✓	✓
Internal loop	✓	✓
Bulge	✓	✓
Multi-loop	✓	✓
Exterior loop	✓	✓
Pseudoknot ($i < k < j < l$)	✓	×
Kissing hairpin	✓	×
Adjacent-base pair ($j = i + 1$)	×	✓

- *Pseudoknots* are admitted by the SAW because the greedy matching procedure does not impose a nesting check, and excluded by the partition function because the recurrence (13) provides no term that could assemble a crossing pair.
- *Adjacent-base pairs* ($j = i + 1$) are admitted by the partition function via the special $i = j - 1$ branch of the

recursion (with the implementation choice $L_{\min} = 0$), and excluded by the SAW because the matching procedure requires $|i - j| > 1$.

- *Multi-partner monomers*, although geometrically possible (a monomer can have up to four lattice neighbours that are themselves chain monomers), are excluded from both pipelines, the SAW by the greedy “first matched, then removed” rule and the partition function by the recurrence structure.

One consequence matters for Section III-H. The ladder distance l measured in the SAW includes pseudoknotted contacts, while the partition function does not. Because the ratio $r = D/(1 - l)$ is computed entirely on the SAW side, both the bare and the corrected calculations use exactly the same r , and this state-space mismatch enters the comparison only as a known simplification.

G. Helicity

The helicity θ is the equilibrium fraction of paired bases at a given temperature,

$$\theta = \frac{2}{N} \left\langle \sum_{i < j} m_{ij} \right\rangle, \quad (19)$$

where $\langle \cdot \rangle$ denotes the thermal average over secondary structures and $m_{ij} \in \{0, 1\}$ is the entry of the pairing matrix, and the factor of 2 counts each base involved in a pair. Helicity is not computed independently of the free energy. It can be extracted directly from F by differentiation with respect to the base-contact energy parameter ε_0 ,

$$\theta = -\frac{1}{N} \frac{\partial F}{\partial \varepsilon_0}. \quad (20)$$

This identity follows from elementary thermodynamics, since the average of any observable is the derivative of the free energy with respect to the parameter that couples it to the energy. Since ε_0 in (11) multiplies the total count of base pairs in (10), its conjugate observable is the fraction of paired bases, that is, θ . The temperature dependence of θ enters through the partition function $Z(T, \varepsilon_0)$ inside $F = -(1/\beta) \ln Z$. Differentiating in ε_0 at fixed T produces a number that depends on T through the Boltzmann weights of all secondary structures contributing to Z .

In practice the derivative is computed numerically by central differences. The partition-function code is evaluated at three closely spaced values of ε_0 , namely $\varepsilon_0 - \delta$, ε_0 , and $\varepsilon_0 + \delta$, and the helicity at the central value is approximated as

$$\theta \approx -\frac{1}{N} \frac{F(\varepsilon_0 + \delta) - F(\varepsilon_0 - \delta)}{2\delta}. \quad (21)$$

The free-energy values for the three ε_0 points are stored at each (b, α, β) combination, where b indexes the bootstrap iteration, as a four-dimensional array. Helicity is computed in a post-processing step by taking the finite difference (21) and averaging over bootstraps.

H. Coupling geometry to thermodynamics, and the independence assumption

The two simulators of this work, the SAW for chain geometry and the McCaskill recursion for base pairing, communicate only through a single number per geometric realization, namely the ratio $r = D/(1 - l)$. This decomposition rests on a simplification, which we now state in plain terms, identify in three distinct places of the methodology, and discuss as a limitation.

1) *The two subsystems and their coupling*: The chain has two distinct kinds of degrees of freedom. On the geometric side, the variables are the lattice positions of the monomers, which together fix the spatial shape of the chain. On the pairing side, the variables form the base-pairing matrix M of the secondary structure, which fixes which bases are bonded to which.

In a fully consistent model these two sets are tightly coupled. A base pair can only form between two monomers that happen to be spatially close, so the geometry constrains which pairings are admissible. Conversely, every pairing pattern forces a particular geometry, since a stem is a rigid double helix, a hairpin loop bends the chain back on itself, and so on. The true joint state of the chain therefore lives in a much smaller space than the product of the two sets considered separately.

2) *The independence assumption*: The simplification used throughout this work is to ignore this coupling at the level of the sampled distributions. We treat the chain shape and the base-pairing pattern as if they were drawn from two unconnected ensembles, and we let them communicate only through one summary number per geometric realization, the ratio $r = D/(1 - l)$. In statistical-mechanics language, the true joint distribution of geometry and pairing is replaced by the product of two distributions that are sampled separately [8]. The geometric side feeds r into the pairing side as a modulation of the field-coupling parameter α , and nothing flows in the opposite direction. The coupling is therefore one scalar, one direction, and it lives entirely inside the value of α .

3) *Where the assumption enters the methodology*: The assumption is not stated anywhere in the pipeline as a separate hypothesis. It is implicit in the architecture of the methodology, and it enters at three points.

The first is the geometric simulation. The SAW samples chain shapes uniformly over the set of self-avoiding walks of the prescribed length, with no preference for shapes that would be comfortable hosts for many favorable base pairs.

The second is the partition function. The McCaskill recursion sums over every nested pairing pattern of the sequence with no requirement that the paired nucleotides be close in space, and the state-space differences of Section III-F add that the two pipelines operate over slightly different families of structures.

The third is the averaging step in (18), where each SAW realization is treated as carrying equal weight and influences the pairing side only through the scalar r_s . A fully coupled treatment would instead reweight each geometric realization

by the pairing Boltzmann factor of that shape, and we do not perform this reweighting.

In other words, running two unconnected simulators and combining their outputs through one scalar per realization is what the independence assumption looks like in practice.

4) *Implication for the test:* The independence assumption underlies both the bare pipeline (the trivial case $r_s \equiv 1$) and the geometrically corrected pipeline (which uses the simulated distribution of r_s). The comparison made in Section IV is therefore internal to this assumption and cannot, on its own, certify the model against a fully coupled ground truth. The methodological consequences of this are discussed in Section IV.

IV. RESULTS AND DISCUSSION

The SAW pipeline retains 200 valid walks for each run, where a walk is considered valid if it reaches the minimum length $N \geq 100$ before becoming trapped. From this sample we extract the chain height D , the ladder distance l , and the ratio $r = D/(1-l)$ for each walk. The 200 ratios are then handed to the bootstrap procedure of Section III-E for use in the geometrically corrected free-energy calculation.

Geometric structure: Figure 2 presents the scatter plot of D against l across the SAW sample. The prior thermodynamic framework implicitly assumes a simple linear relation $D \propto 1-l$, based on the intuition that chains with more internal contacts must be more compact. The simulation confirms the qualitative direction of this dependence, since higher l does correspond on average to lower D , and chains with denser internal contacts do extend less from the surface. The relationship between the two, however, is clearly non-linear. A cubic polynomial fit captures the binned means substantially better than a linear one, and the spread of points around the trend is broad. The analytical substitution $D \rightarrow (1-l)$ therefore reproduces the sign of the height–density correlation but misses both its functional form and the genuine geometric variability that the simulation reveals.

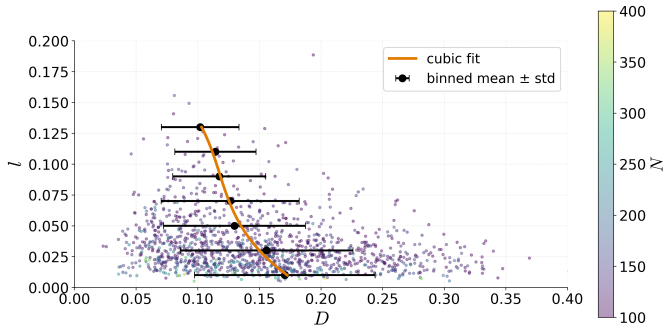


Fig. 2. Scatter plot of average chain height D versus ladder distance l across the SAW sample. Each point corresponds to one self-avoiding walk of length $N \geq 100$ anchored to the surface. The orange curve is a cubic fit to the binned means; black markers show binned means with standard-deviation error bars.

Helicity comparison: Figure 4 compares the helicity curves $\theta(T)$ produced by the bare prior model and the geometrically corrected model, evaluated at the same values of

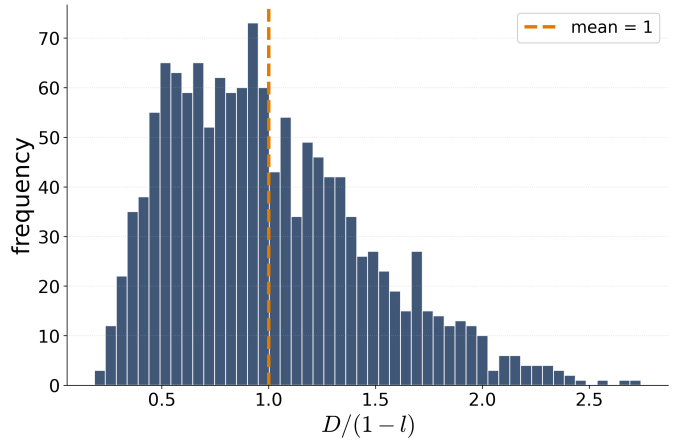


Fig. 3. Empirical distribution of the geometric correction factor $r = D/(1-l)$ across the SAW sample, after normalization to $\langle r \rangle = 1$. The non-zero variance $\text{Var}(r) > 0$ visible in the spread of the histogram is the input to the Taylor-expansion analysis of equations (23) and (24), which traces the helicity gap of Figure 4 to this variance multiplied by the curvature of F in α .

α and T . Recall from Section III-B that α is the strength of the surface field, and that increasing α energetically penalizes folded probe configurations; the precise way in which the penalty depends on the structure is discussed below. At high temperature, in the heat-renaturation regime in which the chain is largely unfolded, the two pipelines produce essentially indistinguishable curves. In this regime the chain occupies extended configurations and the geometric correction has negligible effect.

In the cold regime, the two curves separate. The size of the gap between bare and corrected helicity grows with α , so that stronger field coupling amplifies the difference between the two pipelines. The sign of the gap, however, is itself α -dependent. For moderate α the corrected helicity sits below the bare prediction, indicating that incorporating geometric variability reduces the equilibrium fraction of paired bases. For the largest α shown in Figure 4, the bare model drives the helicity nearly to zero. The corrected curve then sits above it, because there is no room for further suppression below $\theta = 0$ and an asymmetric reweighting around $\langle r \rangle = 1$ can only redistribute the average upward. The shape of $\theta(T)$ for each α , including the field-induced rise of θ with temperature visible at large α that is discussed below, is preserved by the correction. The geometric correction shifts the magnitude of the helicity in the cold regime rather than altering its qualitative temperature dependence.

One point requires care in interpreting the α -dependence of the helicity. In the implemented recursion, after the substitution of the surface term by the ladder-distance proxy, the field coupling enters the statistical weight of a pair through a distance-dependent contribution of the form $\alpha(j-i)$. Thus α should not be read simply as a switch that turns all folding off uniformly. Increasing α penalizes long-range contacts much more strongly than short-range contacts, because the cost grows with the sequence separation $j-i$. The ensemble

can therefore shift away from large loops and compact long-range structures while still retaining, or in some parameter ranges even increasing, the number of short local pairs. Since the helicity counts the total fraction of paired bases without caring about loop length, an increase of helicity at larger α is consistent with suppression of long-range folding, as long as the number of short local pairs grows in compensation. It reflects a redistribution of the secondary-structure ensemble toward shorter contacts rather than a simple monotonic loss of all pairs.

The convergence of the bare and corrected curves at high temperature has a different origin. The Boltzmann weight of a structure is $\exp[-\beta E(M)]$, with $\beta = 1/(k_B T)$. When T is large, β is small, so probability ratios between two structures,

$$\frac{P(M_1)}{P(M_2)} = \exp[-\beta(E(M_1) - E(M_2))], \quad (22)$$

move closer to unity for any fixed energy difference. In this regime the ensemble becomes less selective with respect to energetic details, including the changes induced by α and by the geometric factor r . As a result, varying α or replacing it by αr changes the Boltzmann weights only weakly. This is why the corrected calculation collapses back toward the bare calculation at high temperature, even though the helicity itself need not reach zero over the finite temperature range shown.

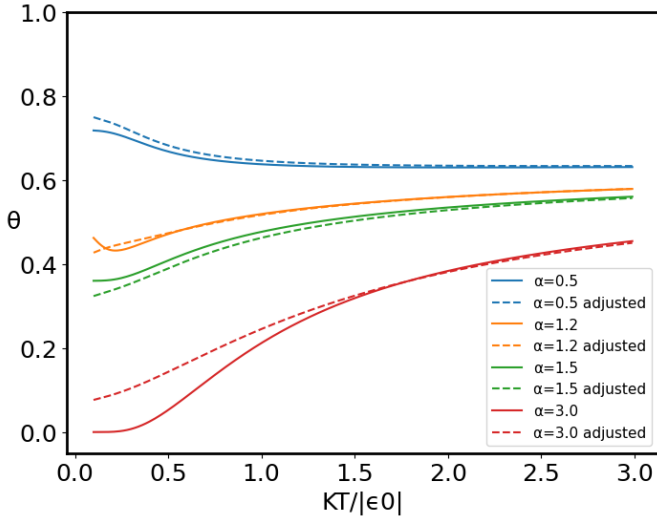


Fig. 4. Helicity $\theta(T)$ for several values of the field-coupling parameter α . Solid lines correspond to the bare prior model, and dashed lines to the geometrically recovered model. The two pipelines agree at high temperature and diverge in the cold regime, with the gap growing with α .

Interpretation via a Taylor expansion: The size and sign of the divergence can be understood from a Taylor expansion of the corrected free energy (18) around the mean $\langle r \rangle = 1$. Writing $F(\alpha r) = F(\alpha) + F'(\alpha) \alpha(r - 1) + \frac{1}{2} F''(\alpha) \alpha^2 (r - 1)^2 + \dots$ and averaging,

$$\langle F(\alpha r) \rangle = F(\alpha) + \frac{1}{2} F''(\alpha) \alpha^2 \text{Var}(r) + \dots, \quad (23)$$

where the linear term in $(r - 1)$ vanishes by the normalization $\langle r \rangle = 1$, so that the first non-zero contribution comes from

the quadratic term in $(r - 1)$ and is linear in the variance $\text{Var}(r) = \langle (r - 1)^2 \rangle$. The leading correction therefore scales as the geometric variance multiplied by the curvature of F in α . Differentiating (23) with respect to ε_0 , since $\theta = -(1/N) \partial F / \partial \varepsilon_0$, yields a corresponding Taylor expansion for the helicity,

$$\langle \theta(\alpha r) \rangle = \theta(\alpha) - \frac{1}{N} \frac{1}{2} \frac{\partial^3 F(\alpha)}{\partial \varepsilon_0 \partial \alpha^2} \alpha^2 \text{Var}(r) + \dots \quad (24)$$

Because ε_0 and α are independent parameters in the recursion, the differentiation in ε_0 commutes with the differentiation in α , and the leading helicity correction inherits the same $\text{Var}(r) \cdot \alpha^2$ scaling as the free-energy correction. This explains why the gap in Figure 4 grows with α and why it is most visible where F is most curved as a function of α , namely in the cold regime. The $\text{Var}(r)$ factor is a fixed property of the SAW geometry, measured once from the simulation, and the temperature-dependent prefactor in front of it determines where the correction is large and where it is invisible.

The same expansion also explains the high-temperature agreement of the two pipelines. Since the r sample is normalized so that $\langle r \rangle = 1$, the linear term in the Taylor series cancels exactly. The first surviving correction is therefore controlled by the curvature of F with respect to α , not by the mean value of the geometric factor. At high temperature the small value of β makes the Boltzmann weights weakly sensitive to energy differences, so the free energy becomes nearly flat or nearly linear as a function of α over the range sampled by αr . In that case $F''(\alpha)$, and correspondingly the mixed derivative $\partial^3 F / (\partial \varepsilon_0 \partial \alpha^2)$ that enters (24), are small. Consequently both the free-energy correction and the helicity correction vanish to leading order, giving

$$\langle F(\alpha r) \rangle \simeq F(\alpha), \quad \langle \theta(\alpha r) \rangle \simeq \theta(\alpha) \quad (25)$$

in the high-temperature regime. The convergence of the curves is therefore not evidence that r has become exactly one for each walk; it is evidence that, in this regime, the thermodynamic observables are insensitive to the variance of r .

Practical implication for biosensors: For biosensor design, the disagreement in the cold regime has practical content. The folding state of the probe directly affects its accessibility for hybridization with the target. If geometric variability changes the predicted helicity at low temperatures and high fields, the bare framework may misestimate the fraction of folded, and therefore partially inaccessible, probes under those operating conditions. The effect is amplified at large α , suggesting that field optimization in biosensor design should account for geometric variability in addition to sequence-level analytical estimates.

Limitation of the test: As noted in Section III-H4, both pipelines compared in Figure 4 rely on the same independence assumption of Section III-H, and the SAW and partition function admit unequal state spaces (Section III-F). The residual difference between the bare and corrected curves therefore measures only the effect of geometric *variance* under the independence assumption. It does not measure the effect of

true geometry–pairing coupling, and it does not measure the effect of the state-space mismatch (pseudoknots in the SAW, $L_{\min} = 0$ in the partition function). A decisive test would require a joint Monte Carlo procedure, in which both geometric and folding moves are proposed under (10) and the pairing matrix is restricted to pairs that are geometrically realizable in the current shape.

Connection to reentrant melting: The non-monotonic helicity behavior identified in [2] for the competitive-energy regime of the two-letter model is not visible in our calculations, because we work in the simpler parameter regime where pair energies are uniformly favorable and the chain exhibits only a single high-temperature melting transition. A natural extension of the present pipeline is to repeat the geometric correction across the parameter ranges of [2] to assess whether geometric variability shifts the location and depth of the low-temperature melting branch. Because the low- T branch is a sensitive consequence of competing pair energies, even a modest geometric correction is likely to leave a measurable imprint on it.

Future work: Several extensions follow naturally from the present study. The most immediate is the implementation of a joint Monte Carlo procedure that samples geometry and folding from the true coupled distribution, removing the independence assumption of Section III-H. Beyond this, the SAW simulations can be extended from two to three dimensions to better reflect the geometry of real biosensor surfaces, and to longer chains representative of biologically relevant probe lengths. The geometric correction r can be made temperature-dependent by reweighting the SAW ensemble against a low-resolution folding score, which would partially restore the coupling neglected by the assumption while remaining computationally tractable. Finally, the methodology can be coupled to brush-electrostatics models of DNA biosensors [4], allowing direct comparison with experimental data on hybridization efficiency.

V. CONCLUSIONS

The free-energy model used in this paper is taken from prior work [1]–[3] together with the biosensor formulation described in the Introduction. The thermodynamic core, namely the partition-function recursion and the analytical substitution $D \rightarrow (1 - l)$, is unchanged. The contribution made here is on the geometric side. A self-avoiding random walk simulation provides the empirical per-walk values of D and l , and therefore the distribution of $r = D/(1 - l)$. This ratio multiplicatively corrects the field-coupling parameter α in the partition function and recovers the geometry-dependent value of D that the analytical substitution had to replace.

The helicity curves produced by the bare prior pipeline and the geometrically corrected pipeline agree at high temperature and diverge in the cold regime, with the gap growing with the field-coupling parameter α . The size and sign of the divergence are accounted for by a Taylor expansion of F around the normalized mean $\langle r \rangle = 1$, in which the leading correction is proportional to the geometric variance $\text{Var}(r)$ and to the

curvature of F in α . Our results therefore quantify, within the assumptions of the prior framework, how much of the helicity is sensitive to genuine geometric variability of the chain as opposed to the sequence-level proxy $(1 - l)$.

The pipeline pays for its simplicity in two ways. First, the two simulators admit slightly unequal state spaces, since the SAW includes pseudoknotted contact patterns that the nested partition-function recursion excludes by construction, while the partition function admits adjacent-base pairs that the SAW excludes. Second, the chain geometry and the base pairing are implicitly treated as statistically independent, with this simplification entering at three points (the SAW sampling, the partition-function summation, and the bootstrap averaging). Both the bare and corrected pipelines share these limitations. The comparison between them therefore measures the effect of geometric variability *within* the independence assumption, not the size of any genuine geometry–pairing coupling. Removing the independence assumption, and reconciling the two state spaces, requires a joint Monte Carlo scheme and is the principal next step.

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